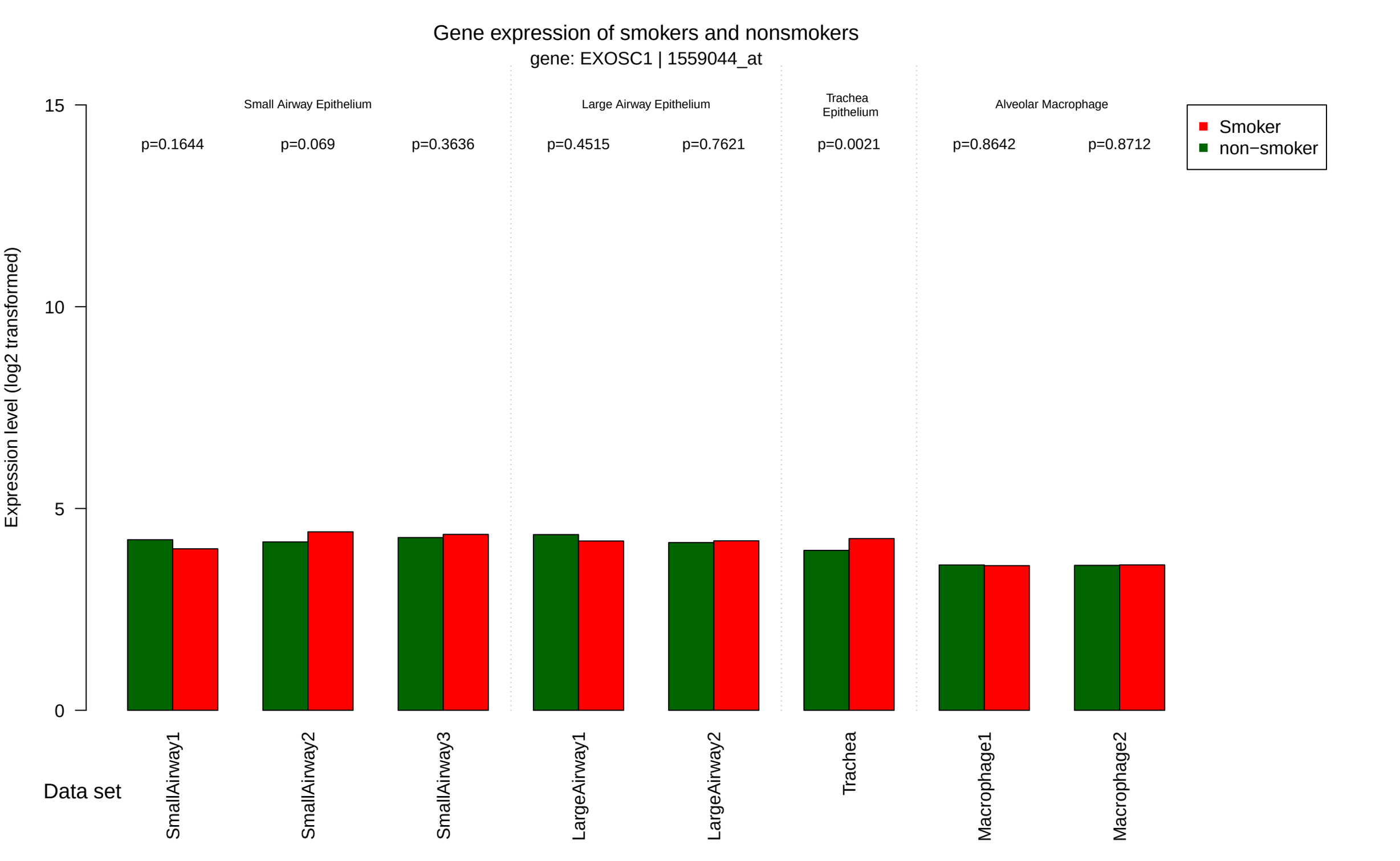
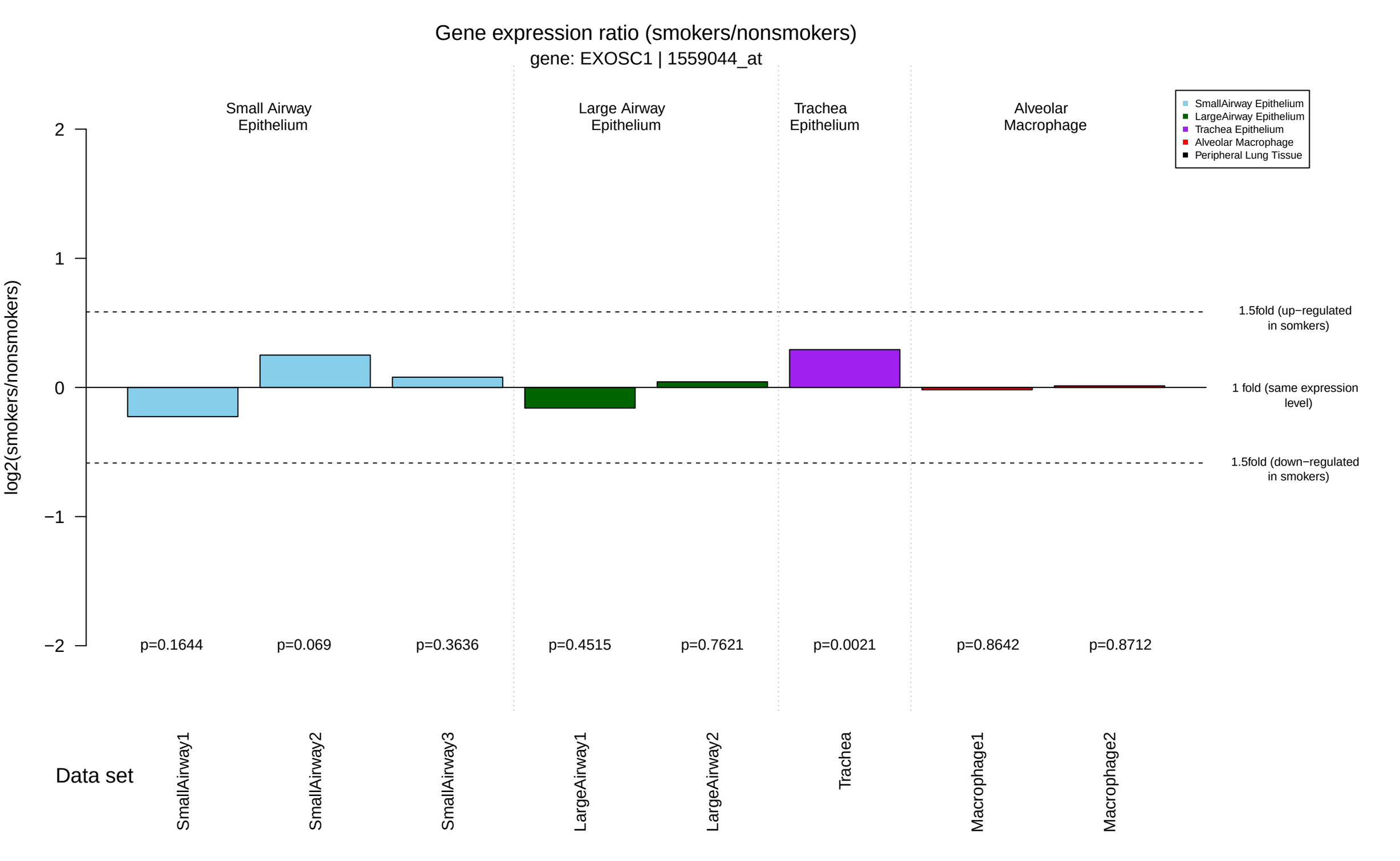
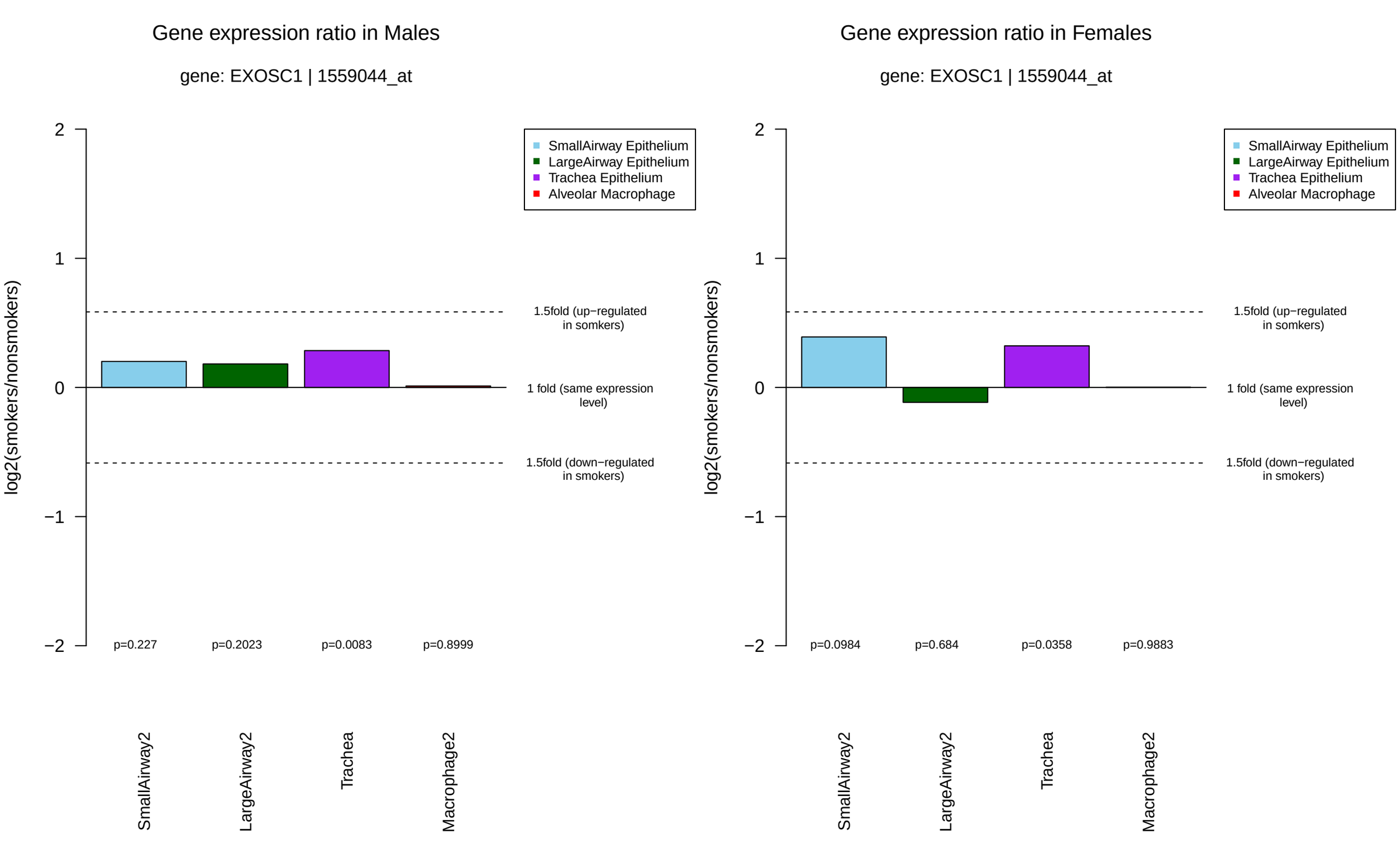
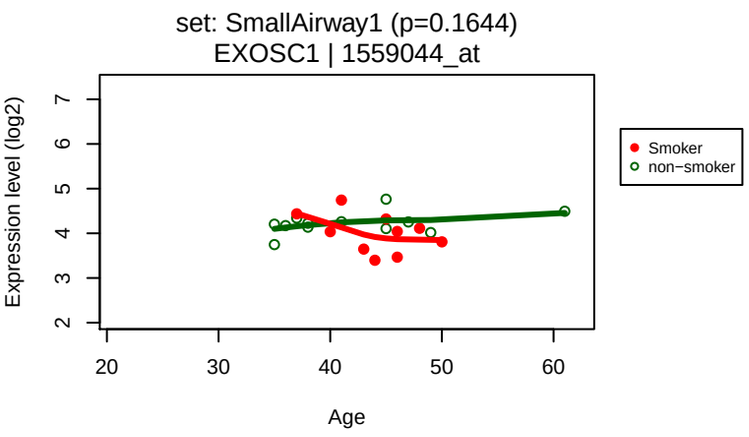
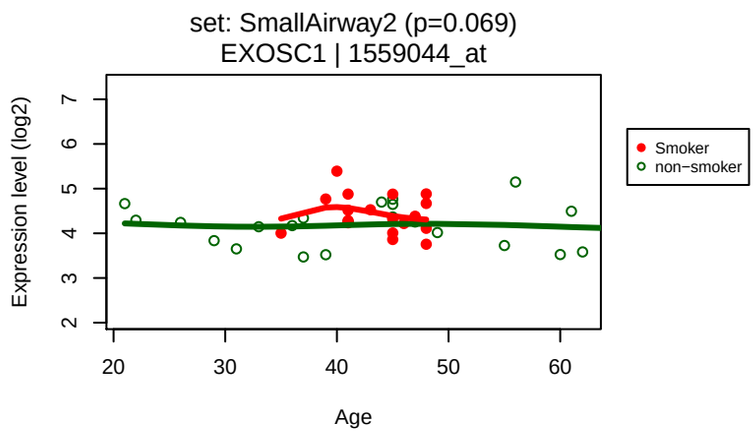
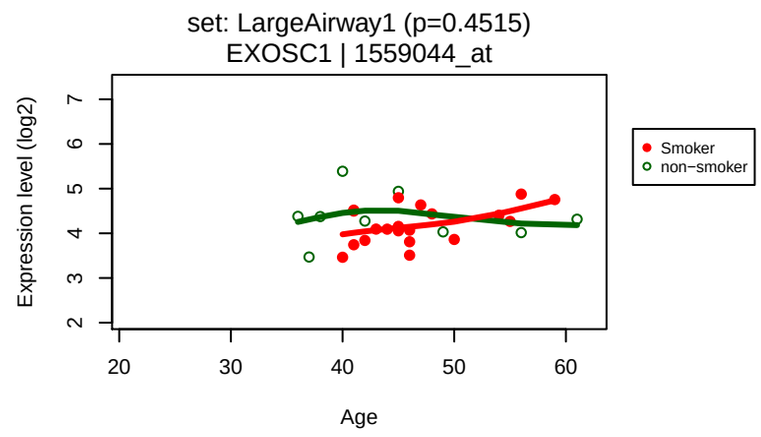

Gene expression of smokers and nonsmokers
gene: EXOSC1 | 1559044_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1644
p=0.069
p=0.3636
p=0.4515
p=0.7621
p=0.0021
p=0.8642
p=0.8712
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

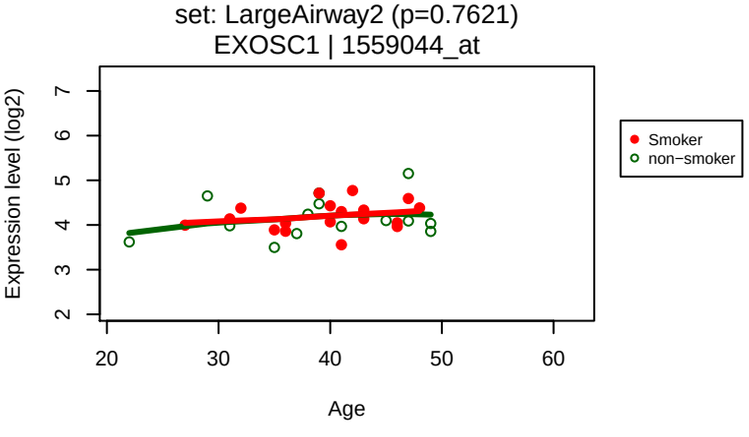

Gene expression ratio (smokers/nonsmokers)
gene: EXOSC1 | 1559044_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1644
p=0.069
p=0.3636
p=0.4515
p=0.7621
p=0.0021
p=0.8642
p=0.8712
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

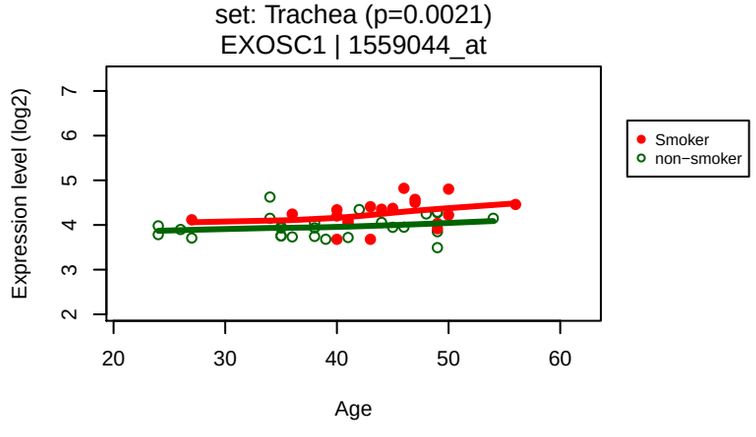

Gene expression ratio in Males
gene: EXOSC1 | 1559044_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.227
p=0.2023
p=0.0083
p=0.8999
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: EXOSC1 | 1559044_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0984
p=0.684
p=0.0358
p=0.9883
SmallAirway2
LargeAirway2
Trachea
Macrophage2

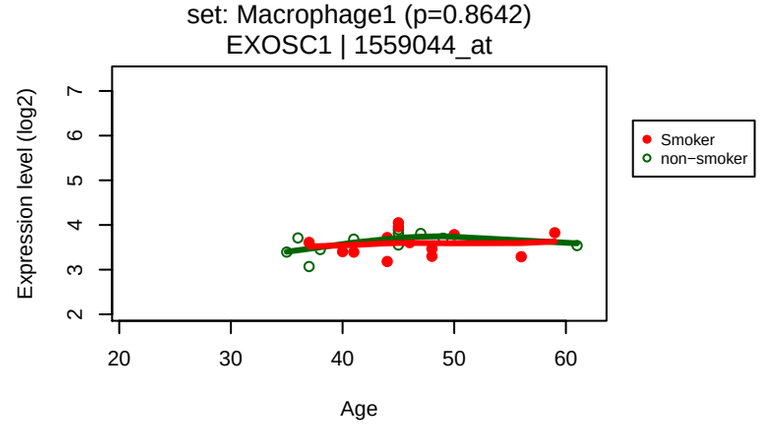

set: SmallAirway1 (p=0.1644)
EXOSC1 | 1559044_at
Expression level (log2)
Age
Smoker
non−smoker

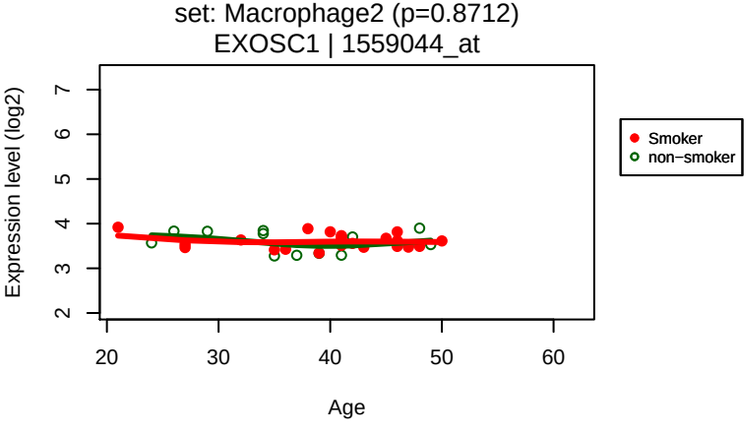

set: SmallAirway2 (p=0.069)
EXOSC1 | 1559044_at
Expression level (log2)
Age
Smoker
non−smoker

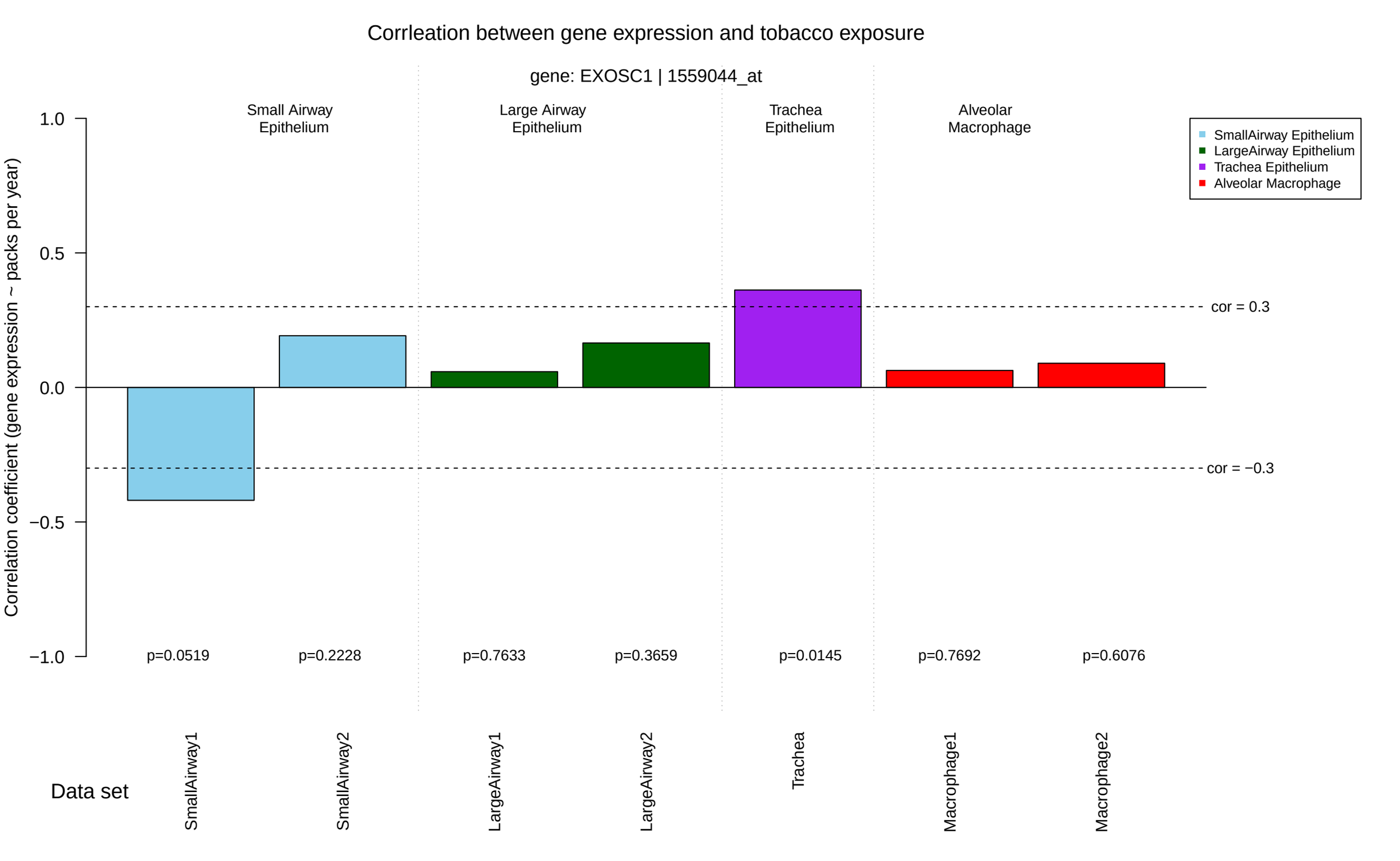

set: LargeAirway1 (p=0.4515)
EXOSC1 | 1559044_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.7621)
EXOSC1 | 1559044_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0021)
EXOSC1 | 1559044_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.8642)
EXOSC1 | 1559044_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.8712)
EXOSC1 | 1559044_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: EXOSC1 | 1559044_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0519
p=0.2228
p=0.7633
p=0.3659
p=0.0145
p=0.7692
p=0.6076
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2